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LAB05: Tiny Transformers on PULP

Davide Nadalini – d.nadalini@unibo.it
Luca Bompani – luca.bompani5@unibo.it
Alberto Dequino – alberto.dequino@unibo.it
Francesco Conti – f.conti@unibo.it

Today Virtuale is down!

To access lab material:

1) Search for: https://github.com/EEESlab/APAI25-LAB05-TinyTransformers
2) Go to the “docs/” folder
3) There, you can find the assignment and the slides!

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Objective of the Class

Intro: Tiny Transformers on PULP

Tasks: execution of a TinyLlama on PULP

- Attention block
- KV cache
- Autoregressive text generation
- Positional encoding
- Tiny story generation with TinyLlama

Deadline:

Nov 7st 2025

Programming Language: C

Lab duration: 3h

<u>The class is meant to be interactive: coding together and on your own!</u>

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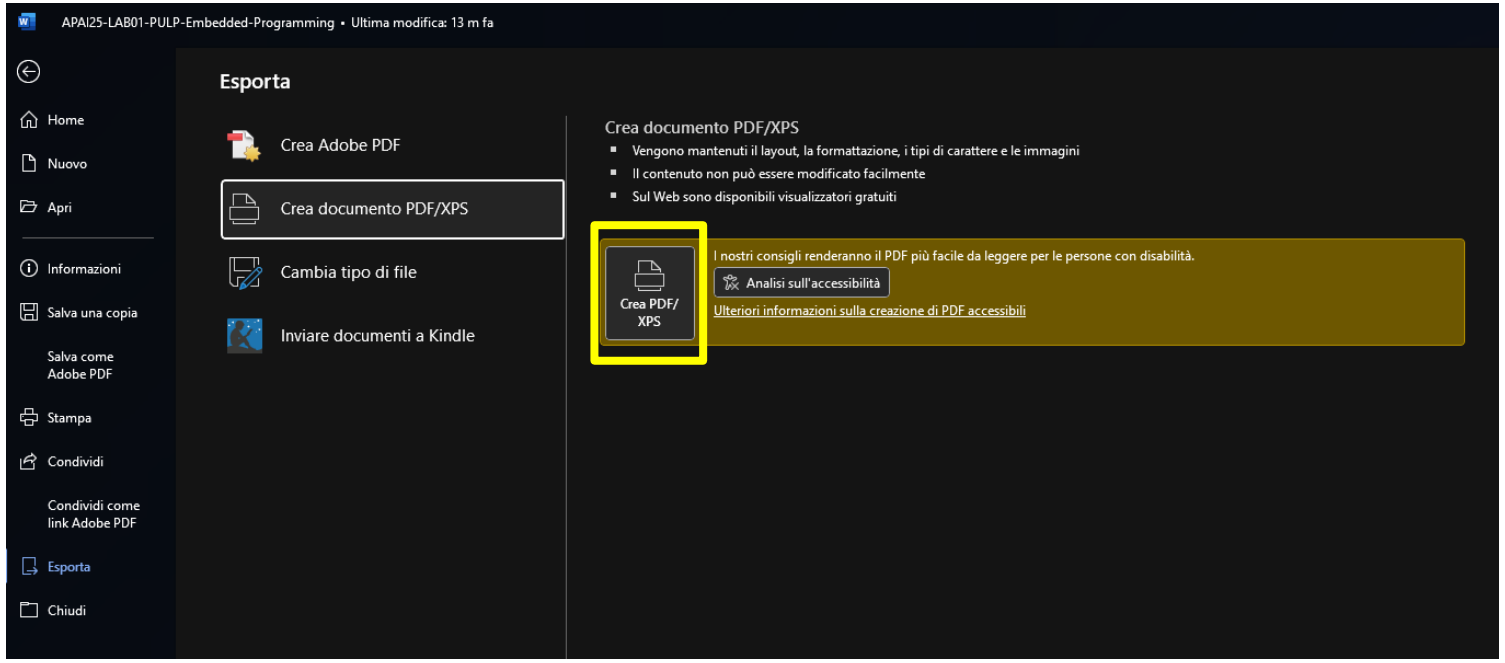
How to deliver the Assignment

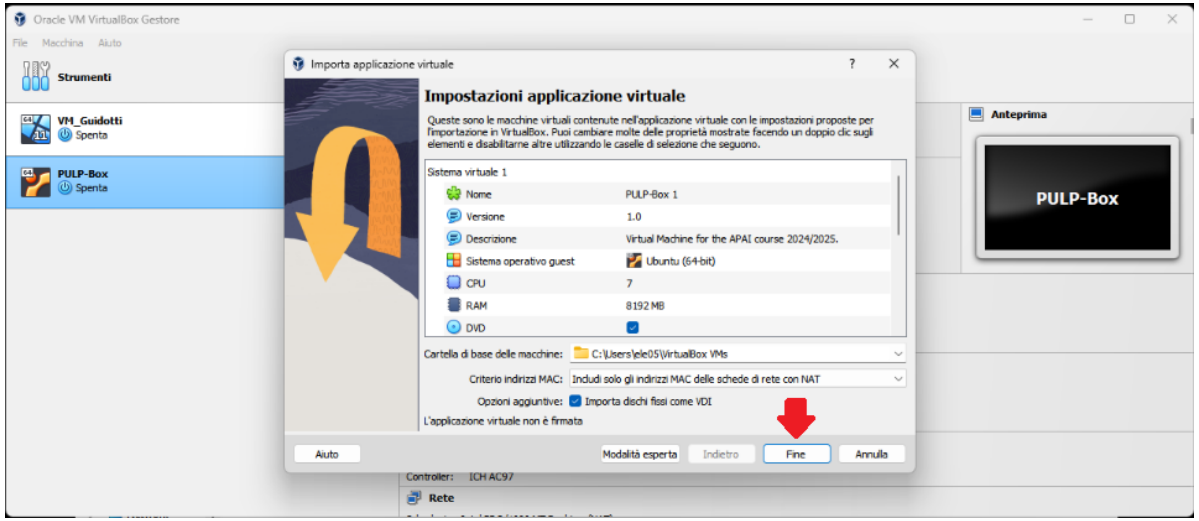
You will deliver ONLY the PDF assignment, no code

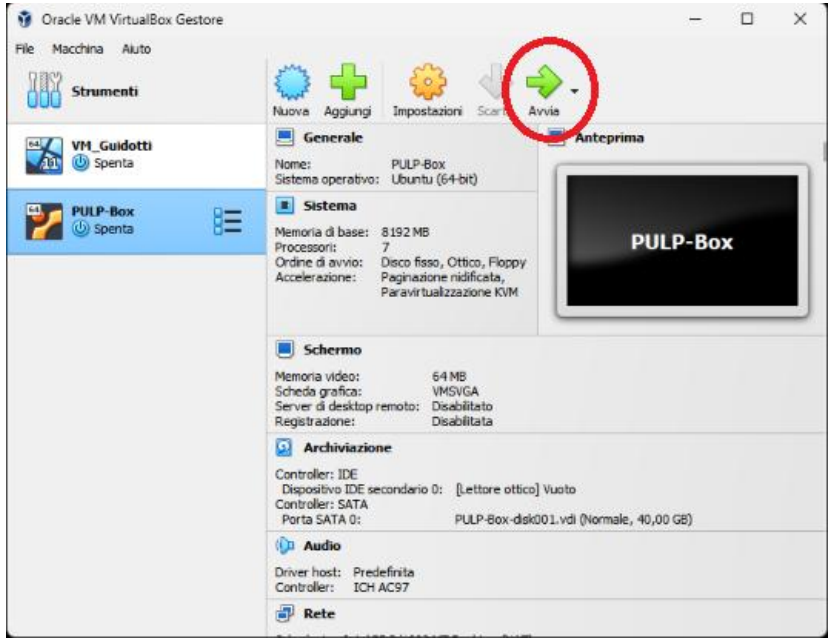
- Download the assignment file from Virtuale.
- Fill the results required by the assignment.
- Export to pdf format.
- Rename the file to: LAB<number_of_the_lesson>_APAI_<your_name>.pdf
- Use Virtuale platform to load ONLY your .pdf file



Opening the VM and VSCode

1. On the lab's PCs, open the file explorer and go to This PC, C:/VM_APAI/
2. Double click on PULP-box.ova
3. VirtualBox opens, just click on "Fine"
4. Wait for the VM to be imported
5. Open the VM with "Avvia"





Password is 'pulp'

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Opening the Docker with VSCode

1. Open a terminal (right click – open a new terminal)
2. From the terminal, open VSCode in the folder of the Docker

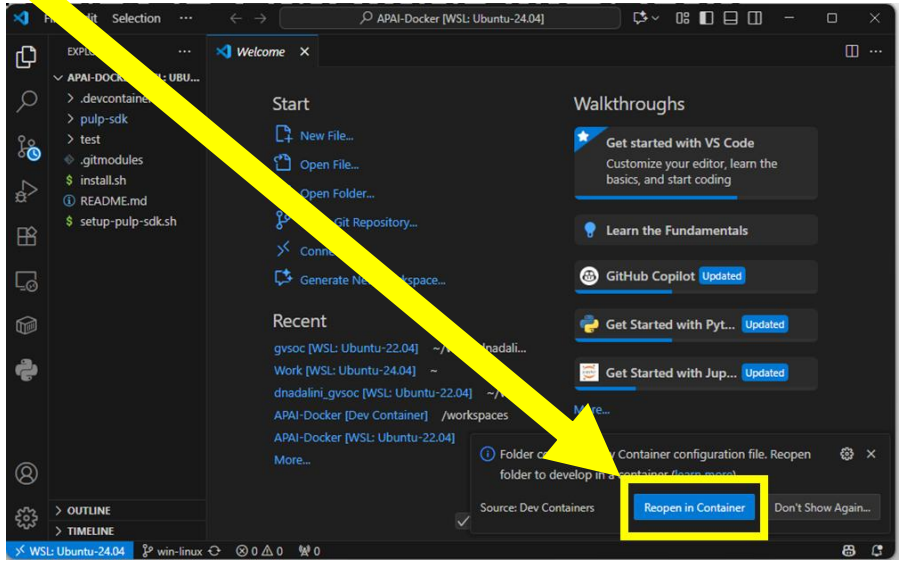
```
$ cd APAI-Docker
$ code .
```

1. Reopen the APAI-Docker folder in VSCode (click on “Reopen in container”)

1. Now you can use the integrated terminal (open with CTRL+J) to run your applications!

IMPORTANT: every time you open a new terminal to work on PULP, launch

```
$ source setup-pulp-sdk.sh
```



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Getting Started:

IMPORTANT: activate the pulp-sdk module file <u>every</u> time a new shell is open.

```
$ source setup-pulp-sdk.sh
```

HOW TO RUN THE CODE:

```
$ git clone https://github.com/EEESlab/APAI25-LAB05-TinyTransformers
$ cd APAI25-LAB05-TinyTransformers
$ cd <folder_you_want>
$ make clean all run
```

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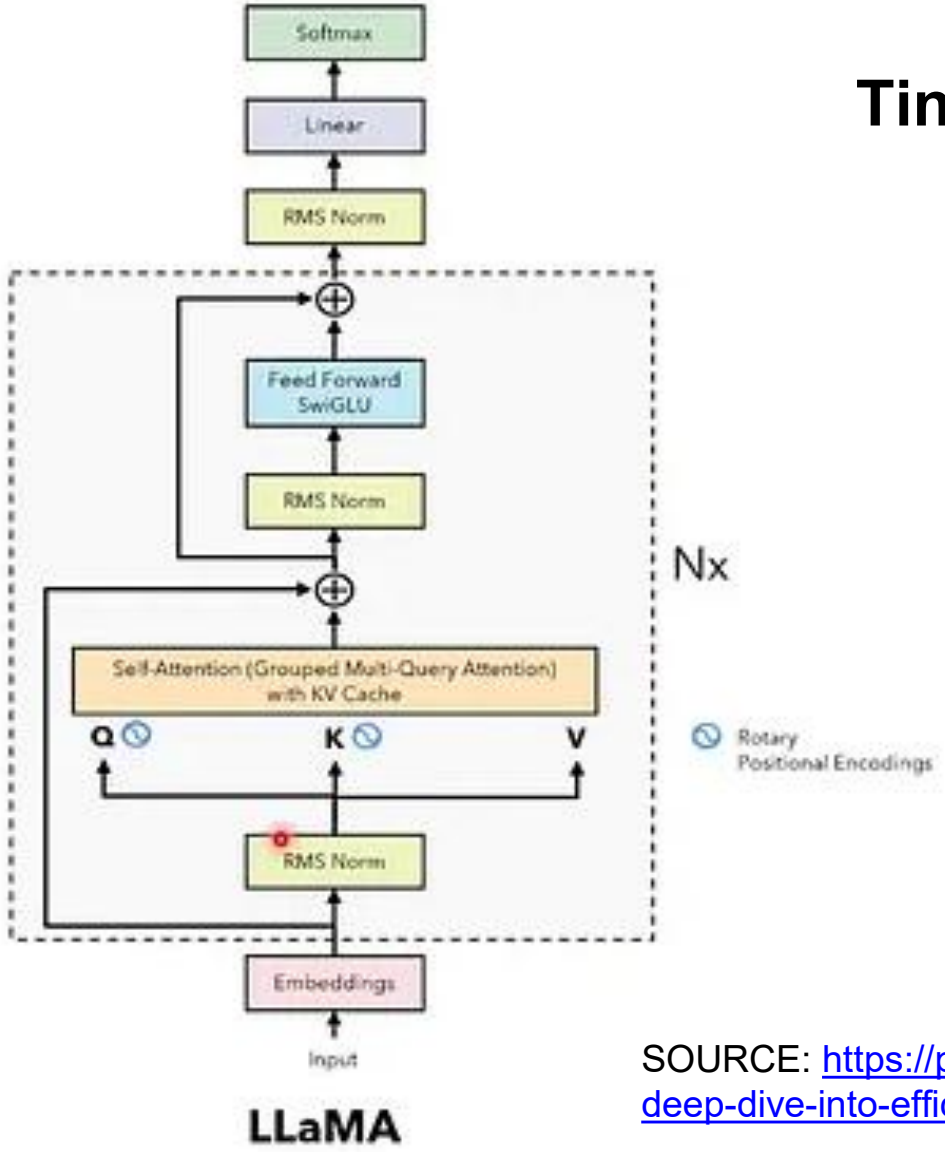
TinyLlama

TinyLlama



TinyLlama: tiny language model based on a Transformer architecture

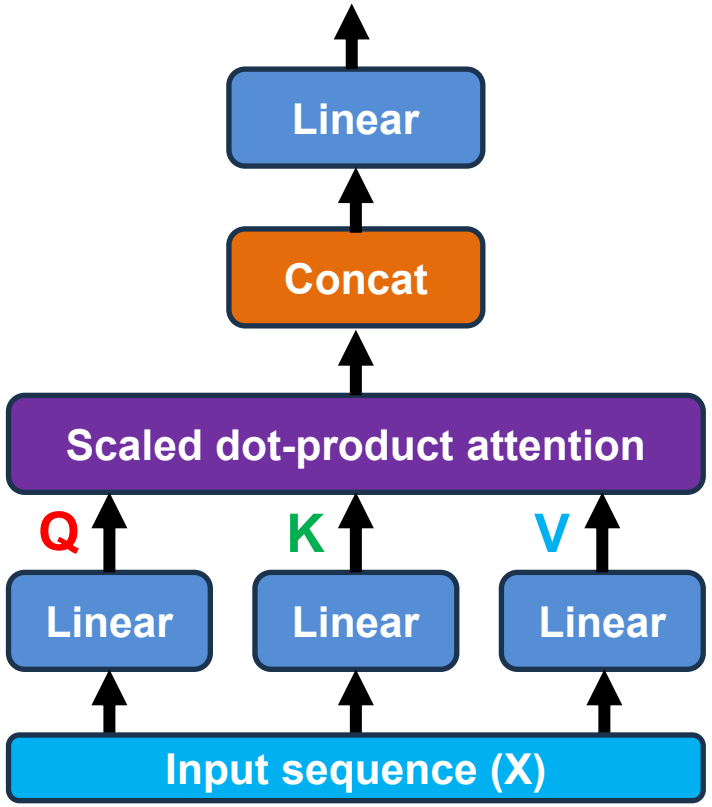
Decoder-only model:

- No Encoder/Decoder, single architecture
- Autoregressive: outputs and inputs are used for text generation

SOURCE: https://pub.towardsai.net/llama-architecture-a-deep-dive-into-efficiency-and-mathematics-9c95c0e4bf8b

Attention Block: Optimization

Core: Attention Block



Given an (encoded) input sequence X, the attention block computes:

Queries: $Q = XW^Q$

Keys: $K = XW^K$

Values: $V = XW^K$

Which are MatMuls!

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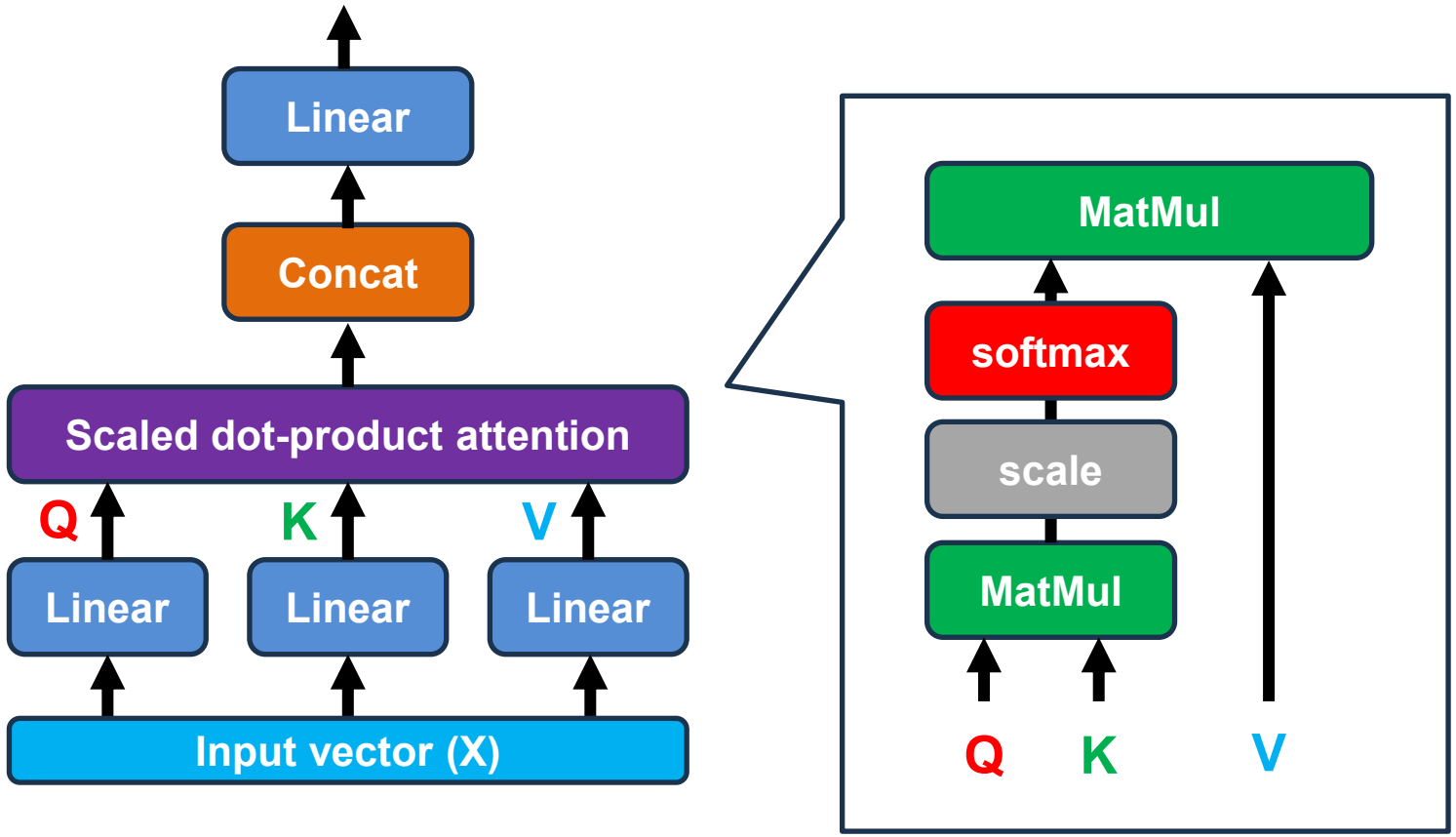
Attention Block: Optimization

Core: Attention Block

$$Attention(Q, K, V) = softmax\left(\frac{QK^t}{\sqrt{d_k}}\right)V$$



The attention block contains the softmax operation, which has exponentials in it

We use FP16 instead of INT8/INT16

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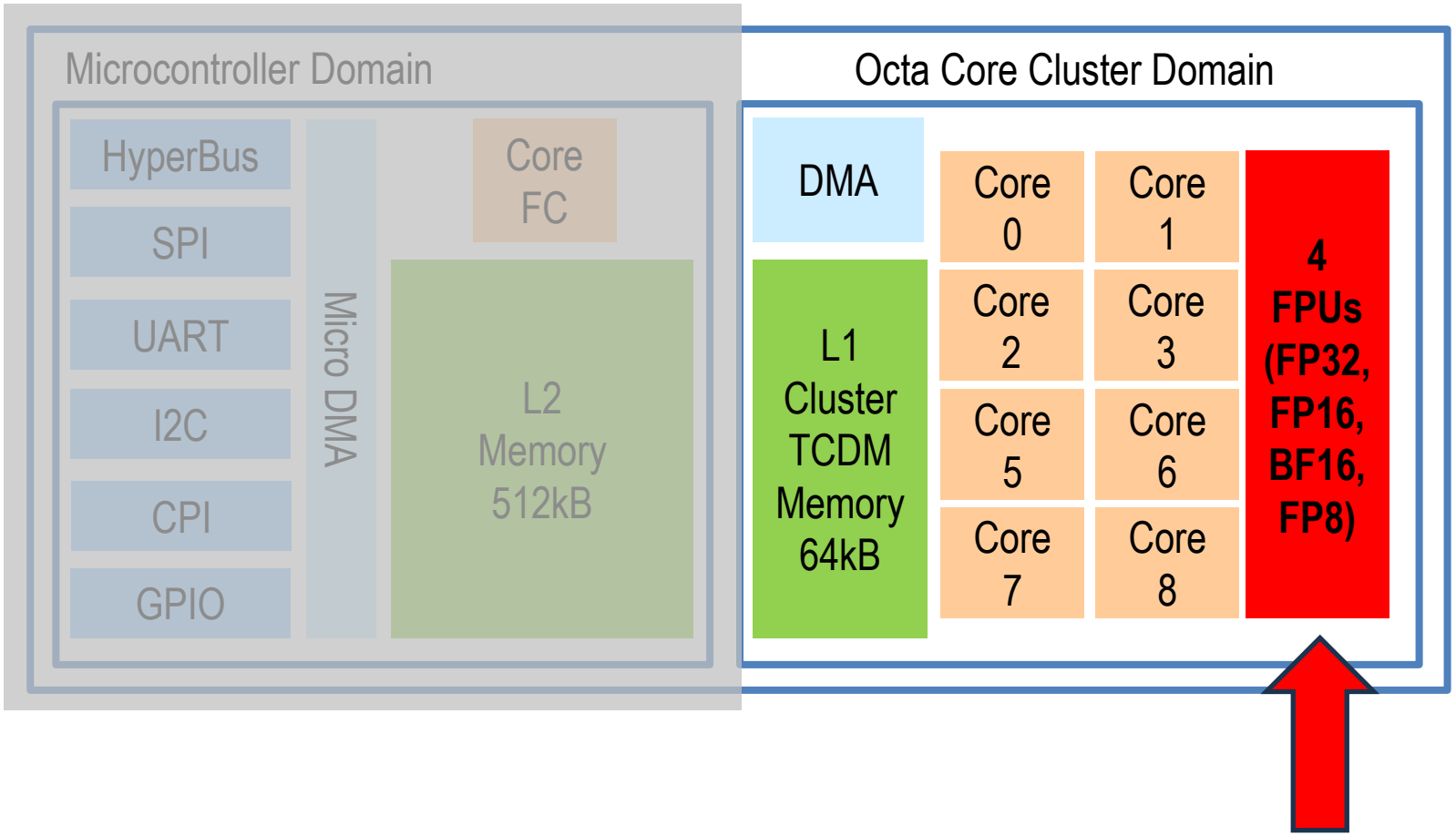
PULP FPUs



However, the PULP Cluster has a mixed-precision floating-point unit (FPU) which can support both FP32, FP16, BF16 and more!

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Mini-Floats

Sign	Exponent	Mantissa
1 b	8 bits	23 bits

Floating-Point 32 (FP32)
IEEE 754 standard (see lecture on Quantization)

Mini-Floats: reduce precision for SIMD and 0.5x memoy usage!!

1 b	5 bits	10 bits

IEEE FP16 – efficient for inference, but not for training

1 b	8 bits	7 bits

BFloat16 – good tradeoff for inference & training!!

In some cases, even 8-bit floating point can be used!!
See, for example: https://arxiv.org/abs/2209.05433

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Using vectorized mini-floats in PULP-SDK

PULP features mini-float data types, as FP16 or BFloat16 as native types in the compiler (e.g., BFloat16 as “float16alt”)

```
typedef float16alt fp16;                                  // FP16 format (float16 or float16alt)
typedef fp16 v2f16 __attribute__((vector_size (4)));      // Vectorized fp16 for SIMD
```

Vectorized FP16 data can be defined as “vectors of 2 elements” for SIMD instructions to exploit the mixed-precision FPU!

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Using vectorized mini-floats in PULP-SDK (cont'd)

```
void vm_T_SIMD (void * void_args)
{
    struct matMul_args_fp16 * args = (struct matMul_args_fp16 *) void_args;
    fp16 * A = args->A;     // In this example, the Output Gradient
    fp16 * B = args->B;     // In this example, the Weight matrix
    fp16 * C = args->C;     // In this example, the Input Gradient
    uint32_t M = args->M;
    uint32_t K = args->K;

    for (int j = 0; j < M; j++)
    {
        v2f16 temp = (v2f16) {0, 0};
        for (int k = 0; k < K; k+=2)
        {
            // Load both adjacent elements for A and B
            v2f16 Av = *((v2f16*) &A[k]);
            v2f16 Bv = *((v2f16*) &B[j*K + k]);
            temp += Av * Bv;
        }
        C[j] = temp[0] + temp[1];
    }
}
```

Initialize a FP16 vector of 2 elements

Load FP16 vectors

Multiply FP16 vectors with SIMD

Vector types (v2f16) can also be used to execute SIMD instruction in a Vector Matrix multiplication, for example!

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